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Lockdown and restaurant closures: evidence from large-scale data in China

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This study examines the impact of the COVID-19 lockdown on China's restaurant industry, a critical contributor to national GDP and employment. Using a large-scale dataset of 14,488,951 restaurant-year observations covering 5,560,345 unique restaurants across 301 cities (2020–2023), we employ the Cox proportional hazard model to examine how lockdown influences restaurant closure. We find that each additional 12 days of local lockdown increases the closure risk by 12.7%. While most restaurants face elevated risks, those with higher star ratings are more resilient. Chain restaurants, older establishments, higher-priced venues, and those offering unique cuisines or located near commercial hubs and transit stations are less likely to close. In contrast, newer, independent, lower-priced restaurants, especially those offering common cuisines, providing delivery, or located in less accessible areas, are more vulnerable. These findings highlight the uneven impact of lockdowns across restaurant types and locations and point to the key factors that support restaurant resilience during disruptions.

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Introduction

The outbreak of the COVID-19 pandemic cast a long shadow over the globe, most notably through the extensive containment measures (Flaxman et al., 2020). From complete lockdown to more targeted restrictions, governments navigated the delicate balance between public health and economic growth. Lockdown measures, while essential for controlling the spread of the virus (Chinazzi et al. 2020; Kraemer et al. 2020; Lai et al. 2020; Schlosser et al. 2020; Wellenius et al. 2021), had significant implications for societies and economies globally (Chetty et al. 2024; Park et al. 2023; Sun et al. 2024). A growing body of literature has examined the economic repercussions triggered by the COVID-19 containment measures, including the impact on global supply chains (Aum et al. 2020; Bonadio et al. 2021), unemployment rates (Coibion et al. 2020; Lozano Rojas et al. 2020), financial markets (Aggarwal et al. 2021; Scherf et al. 2022), and other economic indicators (Bonaccorsi et al. 2020; Witteveen and Velthorst 2020).

The restaurant industry, which relies heavily on in-person interaction and plays a significant role in the national economy, was severely affected by COVID-19 containment measures. Although existing research has documented the operational challenges faced by restaurants during lockdown (Bartik et al. 2020; Gupta et al. 2021; Hirokawa et al. 2023; Sardar et al. 2022), limited attention has been given to quantifying their impact on business survival. Resilient restaurants are those that withstand prolonged disruption and continue operating despite financial, logistical, and public health pressures. This study addresses this gap by examining the relationship between lockdown measures and restaurant closures, providing insights into the industry's vulnerabilities and resilience under sustained disruption.

Utilizing the most extensive dataset on Chinese restaurants, this study investigates the long-term impact of lockdown measures on restaurant closures and probes into factors associated with restaurant resilience. We collected restaurant data from 2020 to 2023 in 300 major cities from the Dianping platform, the largest independent consumer review site in China, comparable to Yelp in the United States (Wu et al. 2015). This four-year timeframe spans the full course of the pandemic and captures multiple rounds of lockdowns implemented in the nation, providing a comprehensive understanding of their implications. Our study operates two levels of analysis: city-level and restaurant-level. At the city level, we assess the macroeconomic impact of lockdowns on the restaurant sector by aggregating data from over 300 cities on the duration and extent of lockdowns, sourced from an official Chinese government website.

Diving into the restaurant-level analysis, we analyze 14,488,951 restaurant-year observations to examine how local lockdowns impact business survival. Using a Cox proportional hazard model, we find that each additional 12 days (one standard deviation) of lockdown in a restaurant's vicinity increases its closure risk by 12.7%. While most restaurants face increased closure risks due to lockdown, those with higher star ratings tend to be more resilient. Chain restaurants, older establishments, and higher-priced venues are generally less likely to close. In contrast, newer, independent, and lower-priced restaurants, especially those offering common cuisines or located in less accessible areas, are more vulnerable. These findings highlight the varied impact of lockdown across restaurant types and locations and point to factors that can help restaurants remain resilient during disruptions.

This study makes significant contributions to the literature on pandemic-induced business disruptions. First, we present comprehensive and long-term evidence on the economic costs of lockdowns, addressing limitations in the temporal and geographic scope of prior studies (Baker et al. 2020; Bonaccorsi et al. 2020;

Bonadio et al. 2021; Coibion et al. 2020; Lozano Rojas et al. 2020; Witteveen and Velthorst 2020). For example, Chen et al. (2025) examined the impact of lockdowns on truck flows across 315 Chinese cities, but their analysis concluded in late 2021. In contrast, our study leverages four years of panel data from 301 Chinese cities, spanning the entirety of China's lockdown period. This dataset enables us to quantify the effects of both the duration and geographic scope of lockdowns on business closures and openings, offering a macroeconomic perspective on the cumulative impact of prolonged and widespread restrictions on economic activity.

Second, we provide in-depth insights into one of the most COVID-vulnerable industries: the service sector. This sector experienced exceptional challenges during lockdowns, with sharp declines in demand and substantial revenue losses (Bartik et al. 2020; Fairlie 2020; Yang et al. 2020). Among service industries, restaurants were particularly hard-hit, as surveys consistently highlight their struggles during extended lockdowns (Bartik et al. 2020; Brizek et al. 2021). Our analysis tracks business survival at the local level, capturing the direct effects of lockdowns on restaurants in their immediate vicinity. This approach enables us to assess both the magnitude and persistence of lockdown-induced disruptions, shedding light on the broader implications for service industries.

Third, we advance the understanding of firm-level characteristics that enhance resilience during lockdowns. Our dataset, comprising 14 million observations from over 5 million unique restaurants, facilitates granular and systematic analysis across a wide array of business attributes such as chain affiliation, business age, pricing strategies, customer ratings, cuisine type, delivery availability, and geographic location. This breadth and depth allow us to uncover nuanced patterns of resilience that might remain undetected in smaller samples, offering actionable benchmarks for operators navigating prolonged disruptions.

Literature Review

The COVID-19 pandemic has prompted numerous studies on its significant economic outcomes. In response to the pandemic, governments around the world implemented lockdown measures to slow down the virus's spread by limiting movement and social interactions (Gupta et al. 2021). However, these measures also came with serious economic costs, increasing unemployment (Aum et al. 2020; Baek et al. 2021), disrupting supply chains (Baqae and Farhi 2022; Bodenstein et al. 2022; Bonadio et al. 2021), and hindering economic activity (Chetty et al. 2024). In the U.S., for instance, 60% of job losses between March 12 and April 12, 2020, were a result of social distancing measures (Gupta et al. 2023). Similar effects were observed in China. It was documented that the 76-day COVID-19 lockdown in Hubei had a severe negative impact on the province's GDP in the first quarter of 2020 (Ke and Hsiao, 2022). Chen et al. (2025) found that a one-month full-scale lockdown in China reduced truck flows to a lockdown city by 54% based on a gravity model of city-to-city trade, indicating a decrease in real income.

A growing body of research has examined the impact of lockdown on the service sector, such as retailing, entertainment, and hospitality, where face-to-face interactions are essential (Belitski et al. 2022). These sectors experienced layoffs exceeding 50% and suffered greater sales losses than other sectors (Bartik et al. 2020; Fairlie 2020). In particular, Baker et al. (2020) found that government-imposed restrictions had a more severe economic impact in service-oriented economies, contributing to the U.S. stock market's sharp decline during the COVID-19 shock. The restaurant industry, heavily reliant on in-person service, was

among the hardest hit (Sardar et al. 2022; Song et al. 2021; Luo and Xu 2021) because it faced significant financial strain and operational difficulties (Kim et al. 2021b). Yang et al. (2020) found that stay-at-home orders caused a daily decline of 3.25% in restaurant demand. Surveys revealed that 22% of restaurant owners expected to close during the initial two-month period (Fairlie 2020), and survival expectations dropped from 74% to 19% as the projected crisis duration increased from one to six months (Bartik et al. 2020). By May 2020, another survey found that 25% of restaurants had already closed permanently (Brizek et al. 2021).

However, the challenges faced by businesses during lockdown were highly heterogeneous, varying by location, size, and operational characteristics. In terms of the regional difference, urban areas saw bigger increases in firm closures than in rural areas in 2020 Q2 through 2022 Q4 (Mitze and Makkonen 2024), likely due to stricter mobility restrictions and higher exposure to consumer-facing industries. The financial support provided by local governments was also shown to reduce closure rates and facilitate the reopening of dine-in establishments (Oikawa and Onishi 2025). At the firm level, Muzi et al. (2023) found that higher pre-pandemic productivity and maintaining a digital presence (e.g., websites, online sales) supported firm survival, while limited operational experience and burdensome business environments increased the likelihood of exit. For restaurants, factors such as larger size, stronger financial condition, brand strength, and positive online reviews have been found to reduce uncertainty during lockdown (Kim et al. 2021a; Neise et al. 2021; Song et al. 2021; Yang et al. 2021). Kim et al. (2021b), analyzing monthly sales data from 86,507 restaurants in nine Chinese cities, showed that offering delivery, providing discounts, and adjusting service types contributed to business continuity during restrictions. Other studies have highlighted the importance of actions that shape consumer perceptions of safety and quality. For instance, the adoption of contactless features improved customer confidence and boosted repurchase intentions during the pandemic (Jeong et al. 2021). A common limitation of the existing literature is the reliance on short observation periods or restricted samples, with few studies leveraging detailed longitudinal firm-level data that span multiple waves of the pandemic.

Building on prior research, our study advances the understanding of the economic consequences of lockdowns by focusing on the restaurant sector in China and systematically examining the impact of localized lockdowns on restaurant operations. Specifically, we leverage a large-scale dataset comprising over 14 million observations spanning four years to quantify how lockdown measures implemented in the vicinity of restaurants influenced closure outcomes and to identify which types of restaurants remained resilient. This extensive and long-term dataset enables us to provide a more comprehensive picture of business closures and the sustained economic costs associated with lockdown measures.

Data

Restaurant data. We obtained restaurant data for China from Dianping.com, the largest independent consumer review website in China (Wu et al. 2015), similar to Yelp in the United States. Dianping.com covers over 300 cities in China, hosting information on more than 1 million local businesses. With a monthly engagement of over 30 million active visitors, users contribute reviews and ratings to help others make well-informed choices about where to dine. The platform offers search and navigation features for easy business discovery based on location, category, or specific criteria. As the most active review website in China, Dianping.com extensively covers the majority of restaurants. Our

dataset, collected annually from 2019 to 2023 in October, comprises 6,120,815 unique restaurants from over 300 cities. Each restaurant entry includes more than 50 characteristics (e.g., star rating, average price, cuisine type, review count, geographical information, etc.), offering comprehensive information.

We collected web-scraped data on restaurants spanning from 2019 to 2023. The concept of a restaurant's closure is defined relative to its presence across consecutive years: if a restaurant is operational in one year but absent in the following year, it is considered to have closed. Accordingly, we commence our analysis with the pool of restaurants operational in 2019, tracking their continuity into 2020, and apply this methodology progressively through to 2023. Consequently, we derive the survival status of restaurants from 2020 to 2023. To operationalize this, we mark the transition of a restaurant from being present in one year to absent in the subsequent year as a closure for that subsequent year. For example, a restaurant operational in 2019 but not found in 2020 is marked as closed in 2020; similarly, if a restaurant exists in 2020 but disappears in 2021, it is considered closed in 2021. We represent this closure status with a binary indicator ($Closure_{it} = 1$ or 0, where 1 signifies closure, and 0 indicates continued operation). The 2019 restaurant data, serving as our baseline for tracking closures, is excluded from the main analysis, which focuses on the period from 2020 to 2023.

Figure 1 shows restaurant closure rates in different provinces in China from 2020 to 2023. This figure shows the lower restaurant closure rates across Chinese provinces in 2020, partly due to the absence of COVID-19 outbreaks in 2019. However, following the outbreaks in 2020, closure rates increased in the subsequent years of 2021, 2022, and 2023. It also reveals that the western regions exhibit lower closure rates compared to the central areas. Central regions of China, such as Wuhan, are highly urbanized and densely populated (Figure A1 of the SI Appendix shows the average population density of Chinese cities). These areas experience greater human mobility, increasing the risk of virus transmission. As a result, these areas are more likely to experience more frequent lockdown measures and prolonged business restrictions, contributing to higher restaurant closure rates. In contrast, many western regions, such as Tibet, Qinghai, and parts of Xinjiang, have lower population densities and less frequent large-scale outbreaks, allowing restaurants to sustain operations more effectively.

We also introduce a variable to track new restaurant openings ($Entry_{it}$). Employing a similar approach, we use the existence of restaurants in 2019 as a baseline to identify any establishments appearing for the first time in the subsequent year. Thus, a restaurant is classified as a new entry if it makes its initial appearance in our dataset in the year following its baseline comparison, indicating it began operations that year.

Figure 2 illustrates the trends in operational restaurants, closures, and new entries. Figure 2 illustrates a drop in the number of operational restaurants in 2021 and again in 2023, and an increase in closures after 2020. This pattern aligns with fluctuations in revenue within China's catering sector (see SI Appendix, Figure A2 for details). After excluding baseline data from 2019 and observations with missing key variables, our dataset from 2020 to 2023 includes 5,560,345 unique restaurants¹.

Lockdown data. China's response to the COVID-19 pandemic unfolded in two phases: the first stage involved stringent, full-scale lockdown measures, followed by targeted interventions such as risk zone classification. During the early phase (2020), the Chinese government enforced strict measures to curb the virus's spread, exemplified by the full-scale lockdown of entire cities or primary urban districts, such as Wuhan. This lockdown often

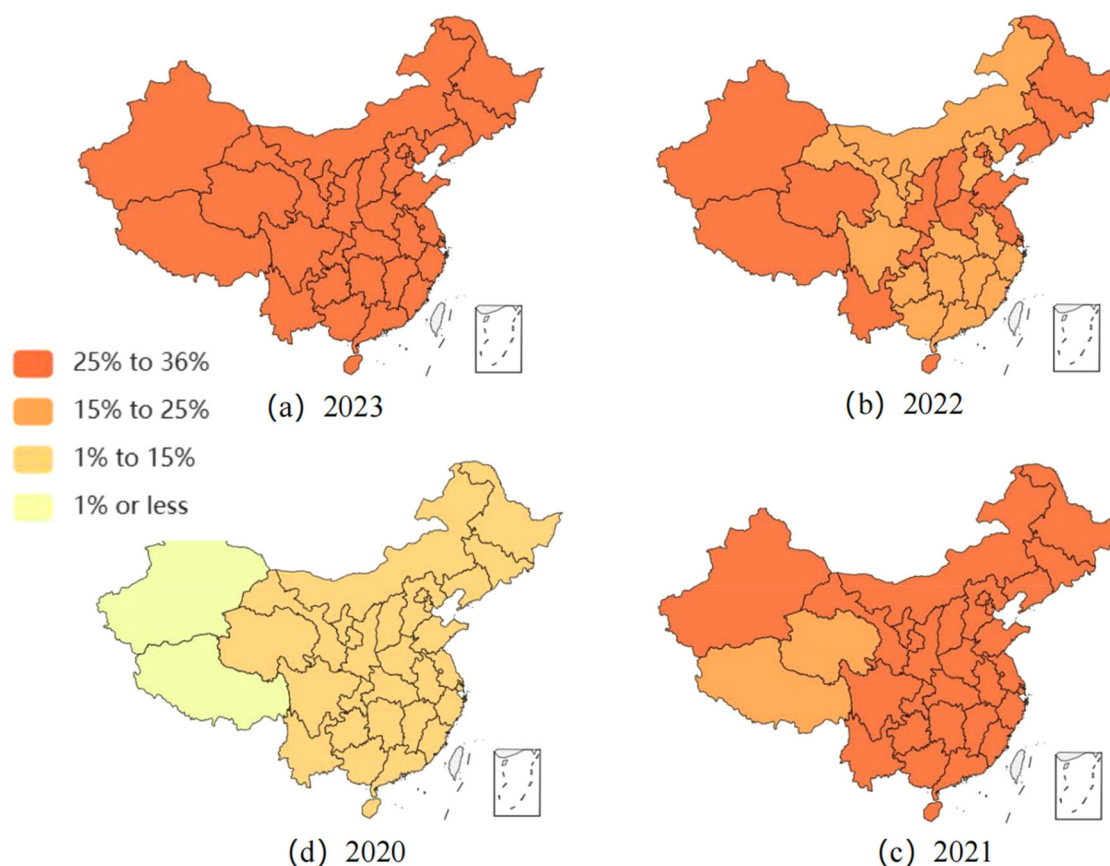


Fig. 1 Restaurant closure rates of different regions in China, 2020–2023. Restaurant closure rates are calculated as the number of closed restaurants in a province divided by the total number of restaurants in the province. Panels **a–d** respectively present restaurant closure rates for 2023, 2022, 2021, and 2020. The figure illustrates a higher restaurant closure rate during 2021, attributed to the 2020 outbreak. Starting in 2021, lockdown measures became more targeted and precise, resulting in a decrease in the closure rate of restaurants in 2022. However, with the widespread impact of China's pandemic on most cities in 2022, more restaurants closed in 2023.

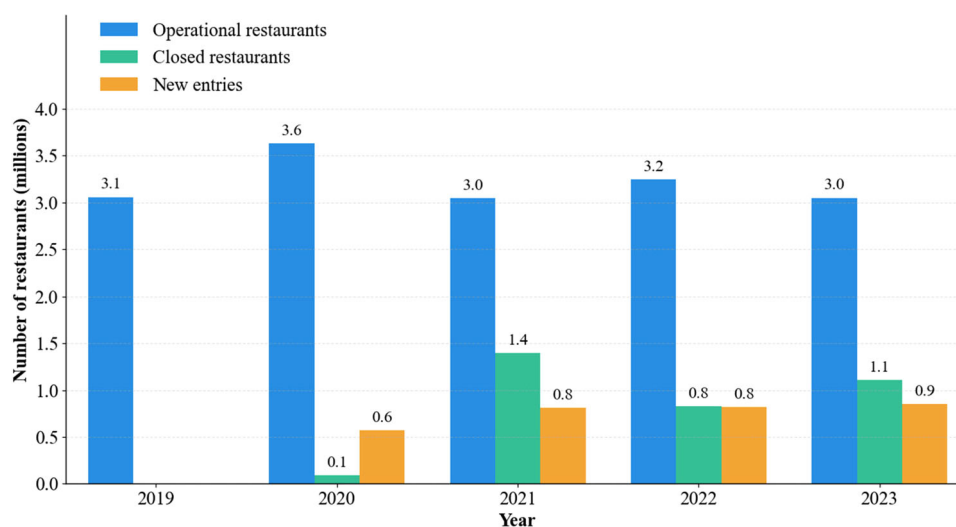


Fig. 2 Trends in restaurant numbers over time. This figure shows the annual changes in the number of operational restaurants, closures, and new entries based on data from the Dianping website.

occurred suddenly and without centralized notification, necessitating reliance on news sources for information. Government announcements, including suspension of all traffic, closed-off management for all residential buildings, and restrictions on

leaving the city, provided critical details (Fang et al. 2020) (see SI Appendix, Figure A3 for details).

Following prior work (Chen et al. 2025), we utilize web scraping techniques to construct a dataset covering the full-scale

lockdown that occurred between January 2020 and December 2020. Specifically, we identify official announcements of lockdown measures by searching for three specific keywords commonly used in such announcements: (1) “closed-off management in all areas,” (2) “traffic controls in all roads,” and (3) “public transport out of service.” Utilizing the Baidu search engine, we search for the year, month, and city name along with these keywords, scraping the first 50 results. We then manually process these web pages, excluding irrelevant pages with inconsistent timing or location.² This dataset includes information about cities that underwent full-scale lockdowns and the number of lockdown days. This procedure identifies 55 cities in which a full-scale lockdown was imposed during 2020.

Since 2021, in response to the ongoing outbreak, the Chinese government established a pandemic management framework, known as the risk zone system, alongside a centralized platform for information dissemination. This initiative has ensured the provision of detailed, timely, and location-specific data regarding lockdown measures. Specifically, the National Health Commission of China classified risk zones into three levels: high-risk zones, medium-risk zones, and low-risk zones. High-risk zones include areas with confirmed cases, asymptomatic carriers, and locations of frequent interaction with high transmission risks. Medium-risk zones encompass areas that have been visited by cases or asymptomatic individuals, presenting potential transmission risks. These zones underwent regular reviews and adjustments based on epidemiological findings, and stringent lockdown measures with only essential operations like residential services and security measures in place, and residents are confined to their homes except for essential purchases. Low-risk zones refer to other areas in the district where the medium and high-risk zones are located. Local governments determined the extent of these zones, which could range from a single address or building to broader regions such as villages, towns, or cities. Given the relatively high scale of medium- and high-risk zones for closures and containment (Chen et al., 2025), we treat these zones as being locked down.

We obtain medium- and high-risk zone data from the aforementioned Chinese government website, which provides daily updates on the count of new confirmed cases, suspected cases, and risk zones in all cities across China (including municipalities and prefecture-level cities, see SI Appendix, Table A1 for details about the Chinese administrative level). The website also daily publishes detailed information about medium- and high-risk zones, including the risk zone count, risk levels (low, medium, and high), and precise addresses of these zones (corresponding provinces, cities, and districts) (detailed in the SI Appendix, Figure A4). We collect daily updated medium- and high-risk zone information from this website covering the period from 2020 to 2023 through web scraping. It is important to note that there were no additional lockdown measures implemented in China in 2023. The final dataset comprises 1,080,947 risk zone-day observations across 327 cities and 1793 districts in China.³

Analysis

City-level analysis. We first conduct a city-level analysis to examine how lockdown has impacted the restaurant industry, with a focus on both the number of closures and new entry restaurants in each city. We assess the extent of lockdown by considering both its duration and geographic scope within each city over the course of a year.

Specifically, we calculate the number of days in a year that a city is under lockdown, capturing the temporal dimension of the measure. We treat a day in a city as a lockdown day if any district

or county within that city was under lockdown (medium- or high-level risk zones) on that day. We calculate the lockdown days attributed to each city for the year 2020 as the days documented in the full-scale lockdown dataset. To account for delayed effects, we lag this measure by one year, denoted as $Lockdown_duration_city_{it-1}$, ensuring that we accurately capture the impact of the lockdown on the restaurant sector. The measure is computed as follows:

$Lockdown_duration_city$ = The number of lockdown days in a city

Concerning the spatial dimension of the lockdown measure, we calculate the number of districts within a city that experienced a lockdown at least once over the course of a year. This metric, denoted as $Lockdown_spatial_city_{it-1}$, indicates how wide lockdown was implemented across different districts of a city. A district is considered to have experienced a lockdown if it either contained at least one designated risk zone or underwent a full-scale lockdown. The higher the value, the more extensive the lockdown coverage across the city’s districts over a year. This measure is computed as follows:

$Lockdown_spatial_city$ = The number of districts under lockdown in a city

After integrating city-level lockdown data with the annual number of restaurant closures and new openings, we obtain 1172 city-year observations across 301 cities in China. Figure 3, illustrated through margin plots incorporating city and year-fixed effects, reveals a pronounced relationship between lockdown measures and restaurant dynamics. Specifically, we observe a significant positive correlation between the lockdown (both the temporal duration and spatial extent) and the number of closed restaurants (Panel A and Panel B) while controlling for the number of new restaurants. Concurrently, there appears to be a negative association between the lockdown and the number of new restaurants (Panel C and Panel D). These findings underscore the reality that increased lockdown severity—both in duration and geographic coverage—elevates the likelihood of restaurant closures while dampening new entries into the market.

Restaurant-level analysis. In our focused analysis at the restaurant level, we aim to precisely understand the impact of lockdown on individual restaurants. To this end, we calculate the days that each restaurant’s location is under lockdown ($Lockdown_duration_restaurant_{it-1}$), framing our measure as follows:

$Lockdown_duration_restaurant$ = The number of lockdown days at the restaurant’s location

We define each restaurant’s location by the county or district it is located in.⁴ We consider a restaurant affected by lockdown measures if its district or county underwent full-scale lockdowns in 2020 or contained medium- or high-risk zones in 2021 and 2022. For simplicity, we refer to both districts and counties as “districts”. Our measure offers a direct assessment of the impact of lockdown on restaurants by determining the days their specific locations are subjected to lockdown, thus affecting their survival odds. Given that restaurants may not immediately be impacted following the initiation of the lockdown, we include a one-year lag for the lockdown in our analysis. This approach allows us to capture the nuanced effects of lockdown duration on restaurants, considering the temporal delay in the manifestation of these impacts.

Integrating this measure with detailed restaurant data, we assemble a dataset consisting of 14,488,951 restaurant-year observations. These observations span 5,560,345 distinct restaurants, providing a rich foundation for our analysis. SI Appendix,

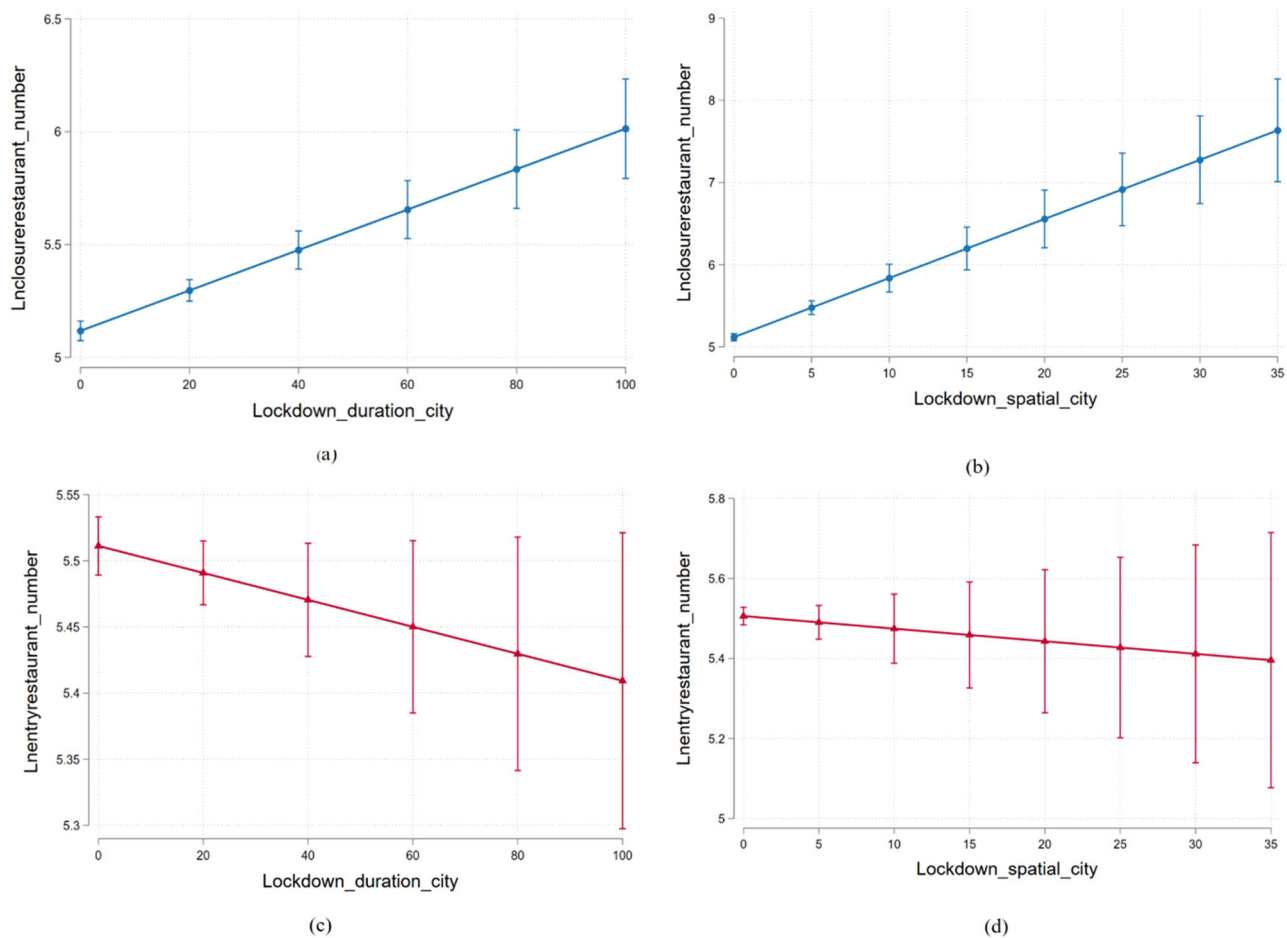


Fig. 3 A margin plot of the lockdown and the number of closed and new entry restaurants. $Lnclosurerestaurant_number_{jt} = \text{Log}(\text{the number of closed restaurants in a given city } j \text{ and year } t) + 1$. $Lnentryrestaurant_number_{jt} = \text{Log}(\text{the number of new restaurants entering markets in a given city } j \text{ and year } t) + 1$. This figure reveals a pronounced relationship between lockdown measures and restaurant dynamics. We observe a significant positive correlation between the lockdown (both the temporal duration and spatial extent) and the number of closed restaurants (**a**, **b**). Concurrently, there appears to be a negative association between the lockdown and the number of new-entry restaurants (**c**, **d**).

Tables A2 and A3 show the distribution of observation frequencies by province and cuisine type.

We assess the hazard of restaurant closure using event history analysis, a method well-suited for discrete events within a defined period. Given our finite observation window, cases with no closure at the window's end are naturally right-censored (Allison 1984). This means we know that these restaurants have not closed, but we do not know exactly when they will close. The Cox proportional hazards model, a form of event history analysis, effectively handles such censored data without excluding it, allowing us to make the best use of all available information (Greene 2003). Additionally, the Cox model is semi-parametric, meaning it does not require a specific assumption about the baseline hazard function. This flexibility is particularly useful when the underlying hazard function is unknown or difficult to model (Cox 1972). By not imposing a fixed form for the baseline hazard, the model enables more accurate, data-driven estimates, which is especially important when analyzing complex events like restaurant closures under uncertain conditions, such as lockdown measures. Previous research has empirically demonstrated the robustness of this model (Chen et al. 2023; Waguespack and Fleming 2009). For these reasons, we use this model to explore the impact of lockdown measures on the likelihood of restaurant closure, while also accounting for any potential heterogeneity

effects.

$$\begin{aligned} \text{Hazard}_{ijt} &= h(t, |, \text{Lockdown_duration_restaurant}_{it-1}, C_{it}, \text{YearFE}_t, \text{CuisineFE}_k) \\ &= h_0^j(t) \exp(\beta_1 \text{Lockdown_duration_restaurant}_{it-1} + \beta_2 C_{it} + \text{YearFE}_t \\ &\quad + \text{CuisineFE}_k) \end{aligned}$$

In this equation, i represents the restaurant, j represents the city, t represents the year, and k represents the cuisine type. The hazard captures the likelihood that a restaurant closes at the current time ($Closure_{it}$). $h_0(t)$ is the baseline hazard shared among restaurants within city j . Stratifying the baseline hazard by city allows for location-specific variations, accounting for diverse survival probabilities across geographies. The variable $\text{Lockdown_duration_restaurant}_{it-1}$ measures the days that each restaurant's location is under lockdown. To mitigate the influence of outliers, we apply a winsorization technique to the independent variables, limiting extreme values to the 99th percentile. C refers to the vector of control variables, including *Chain_status*, *Star_group*, *Price_group*, *Commercialcenter*, *Transportation*, *Delivery*, and *Prevalence_group*. Year FE and Cuisine FE denote year and cuisine type fixed effects, enabling the control for common time and cuisine-related effects. To explore moderating effects, we stratify the hazard by moderators and interact each moderator variable with the lockdown duration. All the key

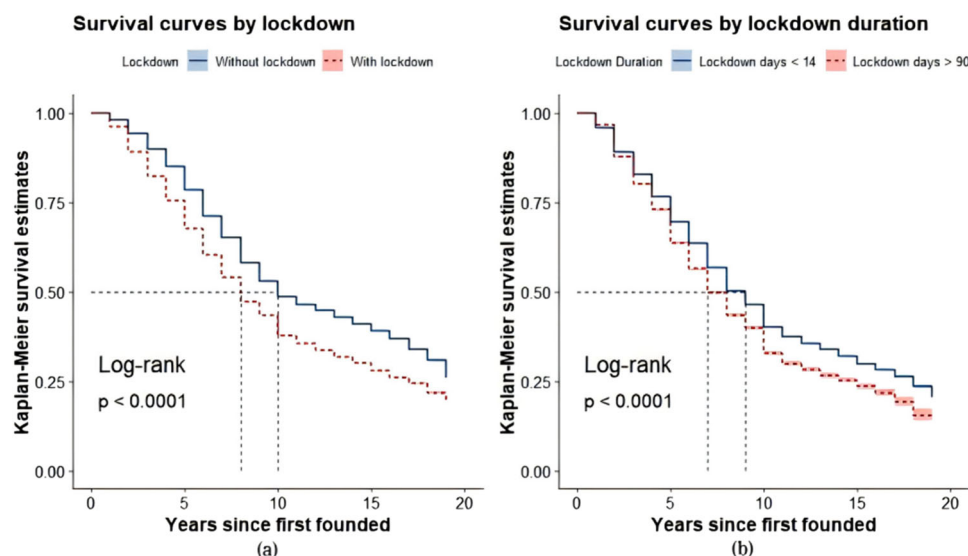


Fig. 4 The rate of restaurant closures based on various lockdown statuses and lockdown duration. Log-rank test result: Log-rank $p < 0.0001$ (indicating the difference between the two curves is significant). (a) illustrates that restaurants facing lockdown measures encounter a lower survival probability, indicating more significant threats. (b) emphasizes that restaurants face a heightened risk of closure as the lockdown duration increases.

variables in the study and their data sources are presented in Table A4 of the SI Appendix.

In Fig. 4a, b, we present a survival probability plot categorized by lockdown status and duration. Panel A shows that restaurants subject to lockdown measures face a significantly increased risk of closure. Panel B illustrates a clear difference for restaurants that undergo a short or an extended period of lockdown. The likelihood of restaurant closure rises with longer lockdown duration. Table A6 in the SI Appendix reveals a positive relationship between lockdown duration and the likelihood of restaurant closure ($b = 0.010$, $p < 0.01$). A one-standard-deviation increase in lockdown duration (approximately 12 days) increases the likelihood of closure by 12.7% ($\exp(0.010 \times 11.940) = 1.127$). We also analyze the effects of lockdown across different restaurant categories to understand their heterogeneity. Table A8 in the SI Appendix shows that the hazard ratio for every cuisine type of restaurant is positive, underscoring the substantial threat to restaurant survival posed by lockdown measures.

Subsequently, we explore factors that could impact a restaurant's resilience. To do so, we stratify the hazard by chain status, age, star rating, average price, proximity to the commercial center and public transportation, availability of delivery, and cuisine prevalence (detailed in Table A4 of the SI Appendix) and interact each of the variables with lockdown duration. We employ z-tests to test whether the coefficients for various groups differed significantly.

A chain restaurant typically operates multiple locations under a single brand or identity with a common name. We define chain status based on the number of related restaurants with a common name in a city. The chain status is categorized as follows: independent (no related restaurants), small chain (up to 50 related restaurants), and large chain (more than 50 related restaurants), reflecting the scale of a restaurant's chain operations ($Chain_status_{it}$). Figure 5 indicates that all independent and chain restaurants face increased hazards of closure during lockdown periods. Examining the differences among different groups, we find that for independent restaurants, every 12 additional days of lockdown (one-standard-deviation increase) results in a 12.7% ($\exp(0.010 \times 11.940) = 1.127$) rise in the likelihood of closure, while for large chain restaurants, the closure likelihood increases by 11.3% ($\exp(0.009 \times 11.940) = 1.113$) (see SI Appendix, Table

A7, column 1). The results from heterogeneity analysis across cuisines further support these findings (see Fig. 6, SI Appendix, Table A8).

Restaurant age is calculated as the difference between the current year and the year a restaurant first appeared on Dianping.com, reflecting its operational history. We categorize restaurant age into three groups: less than 3 years, 3–6 years, and greater than 6 years (Age_group_{it}). Figure 5 shows that all age groups face increased hazards of closure, with younger restaurants being more vulnerable. Comparing different age groups, we find that the hazard ratio decreases as the age of the restaurant increases. Specifically, an additional 12 days of lockdown increases the likelihood of closure by 15.4% for restaurants less than 3 years old ($\exp(0.012 \times 11.940) = 1.154$). For restaurants aged 3–6 years and those older than 6 years, the increased hazard odds are both 6.2% ($\exp(0.005 \times 11.940) = 1.062$) (see SI Appendix, Table A7, column 2). This suggests that restaurants with longer operational histories have likely accumulated more resources to withstand the challenges of lockdown. This trend is consistent across different cuisines (see Fig. 6 and SI Appendix, Table A8).

A restaurant's star rating ranges from 0 to 5, indicating a restaurant's overall reputation and popularity. We split restaurants into four categories based on their star ratings: less than 3.0, 3.0–3.5, 3.5–4.0, and 4.0–5.0 ($Star_group_{it}$). Figure 5 illustrates that as restaurants' rating improves, the impact of lockdown duration on the likelihood of closure decreases significantly. Specifically, our estimate shows that lockdown has a smaller impact on the closure likelihood of higher-rated restaurants, with a 4.9% increase in the likelihood of closure ($\exp(0.004 \times 11.940) = 1.049$) (see SI Appendix, Table A7, column 3). However, for restaurants with the lowest star ratings, lockdown exhibits a substantial impact on closure likelihood ($b = 0.013$, $p < 0.01$) (see SI Appendix, Table A7, column 3). An additional 12 days of lockdown increases the likelihood of closure by 16.8% ($\exp(0.013 \times 11.940) = 1.168$) rise in the likelihood of closure for restaurants less than 3 stars. These results suggest that higher-star restaurants, with their stronger reputation and popularity, show greater resilience during lockdown. Figure 6 further illustrates that for restaurants specializing in drinks, bakeries, cafes, and international cuisine, high-star ratings are particularly important for resilience during lockdown (see SI Appendix, Table A8).

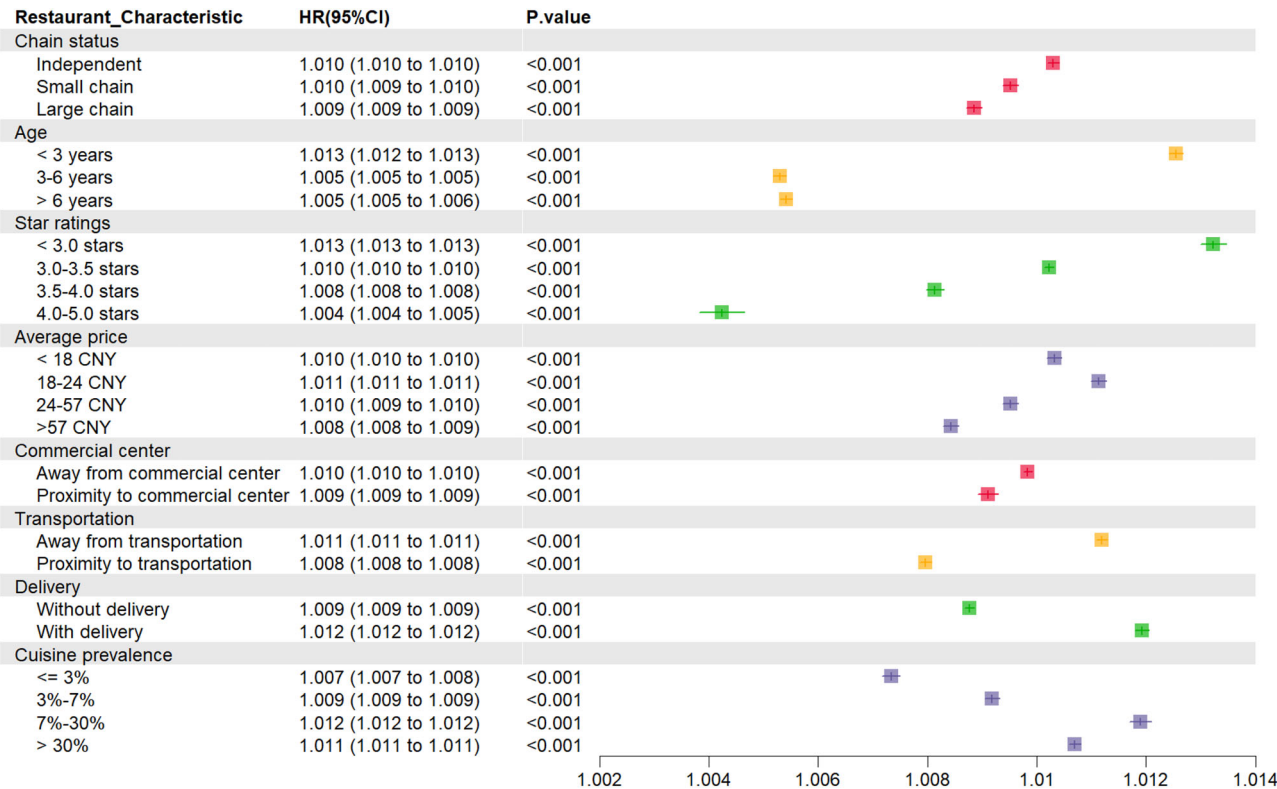


Fig. 5 Restaurant characteristics affecting restaurant resilience during lockdown. This figure shows the hazard ratio for different restaurant groups with 95% confidence intervals (HR (hazard ratio) = exp(b)), where values larger than 1 indicate increased risk and values smaller than 1 indicate decreased risk. High star ratings could contribute to the resilience of restaurants because lockdown has less effects on those restaurants.

The average price per person for each restaurant is categorized based on quartiles: less than 18 CNY (1 USD ≈ 7 CNY), 18–24 CNY, 24–57 CNY, and more than 57 CNY (*Price_group_{it}*). Figure 5 illustrates that all price groups face increased hazards of closure, with low-price groups being more vulnerable. Comparing different price groups, we find that the hazard ratio decreases as the average price of the restaurant increases. Specifically, an additional 12 days of lockdown increases the likelihood of closure by 12.7% ($\exp(0.010 \times 11.940) = 1.127$) for the low-price group, slightly exceeding the increases observed for the high-price group, which are 10.0% ($\exp(0.008 \times 11.940) = 1.100$) (see SI Appendix, Table A7, column 4). Figure 6 demonstrates that a higher average price generally decreases the closure likelihood for most restaurants, but for a minority specializing in hot pot and barbeque, they might slightly increase the closure hazard (see SI Appendix, Table A8).

We assess the proximity to the commercial center (*Commercialcenter_{it}*)⁵ and public transportation (*Transportation_{it}*)⁶, indicating whether restaurants are situated in or near commercial centers like malls and shopping centers and public transportation such as subway stations, bus stops, train stations, or airports. In Fig. 5, it is observed that all groups face increased hazards of closure regardless of their location, while the hazard is relatively lower when the restaurant is situated near public transportation. Specifically, being near a commercial center lowers the hazard by 1.3% ($\exp(0.009 \times 11.940) - \exp(0.010 \times 11.940) = -0.013$) compared to those further away. Similarly, proximity to public transportation reduces the hazard by about 4.0% ($\exp(0.008 \times 11.940) - \exp(0.011 \times 11.940) = -0.040$) compared to restaurants in less accessible locations (see SI Appendix, Table A7, columns 5 and 6). Figure 6 corroborates these findings, as detailed in the SI Appendix, Table A8.

Delivery availability is represented as a binary variable, with 0 indicating no delivery service and 1 indicating the presence of

delivery service (*Delivery_{it}*). Figure 5 illustrates the increased hazards of closure in both groups. Examining the difference between the two groups, we find that restaurants that offer delivery services are more vulnerable than those not. Specifically, every 12 additional days of lockdown increases the closure hazard by 15.5% ($\exp(0.012 \times 11.940) = 1.155$) for restaurants that offer delivery services and by 11.34% ($\exp(0.009 \times 11.940) = 1.1134$) for those that do not (see SI Appendix, Table A7, column 7). Restaurants providing delivery services, particularly those specializing in fast food and noodles, are more susceptible to closure due to lockdown measures, as detailed in Fig. 6 and SI Appendix, Table A8. The provision of delivery services may entail additional expenses, such as fees for third-party delivery platforms or investments in delivery infrastructure. Thus, while delivery services offer a potential revenue stream during restrictions, they also pose a heightened risk of financial burden, leading to a greater vulnerability to closure under prolonged lockdown conditions.

Cuisine prevalence reflects how common a particular type of cuisine is within a market, measured by the percentage of restaurants in a city in a given year that offer a specific cuisine (cuisine K). We hypothesize that higher cuisine prevalence indicates greater competition, potentially exacerbating the impact of lockdown on restaurant closures. To explore this, we classify cuisine prevalence into four groups based on quartiles (*Prevalence_group_{it}*). Our findings indicate that restaurants in all prevalence groups experience an increased hazard of closure during the lockdown, but those in high-prevalence categories are particularly vulnerable. Specifically, restaurants with low cuisine prevalence see an 8.7% increase in closure likelihood for every additional 12 days of lockdown ($\exp(0.007 \times 11.940) = 1.087$), while those in high-prevalence categories face a more pronounced 14.0% increase ($\exp(0.011 \times 11.940) = 1.1404$) (see SI Appendix,

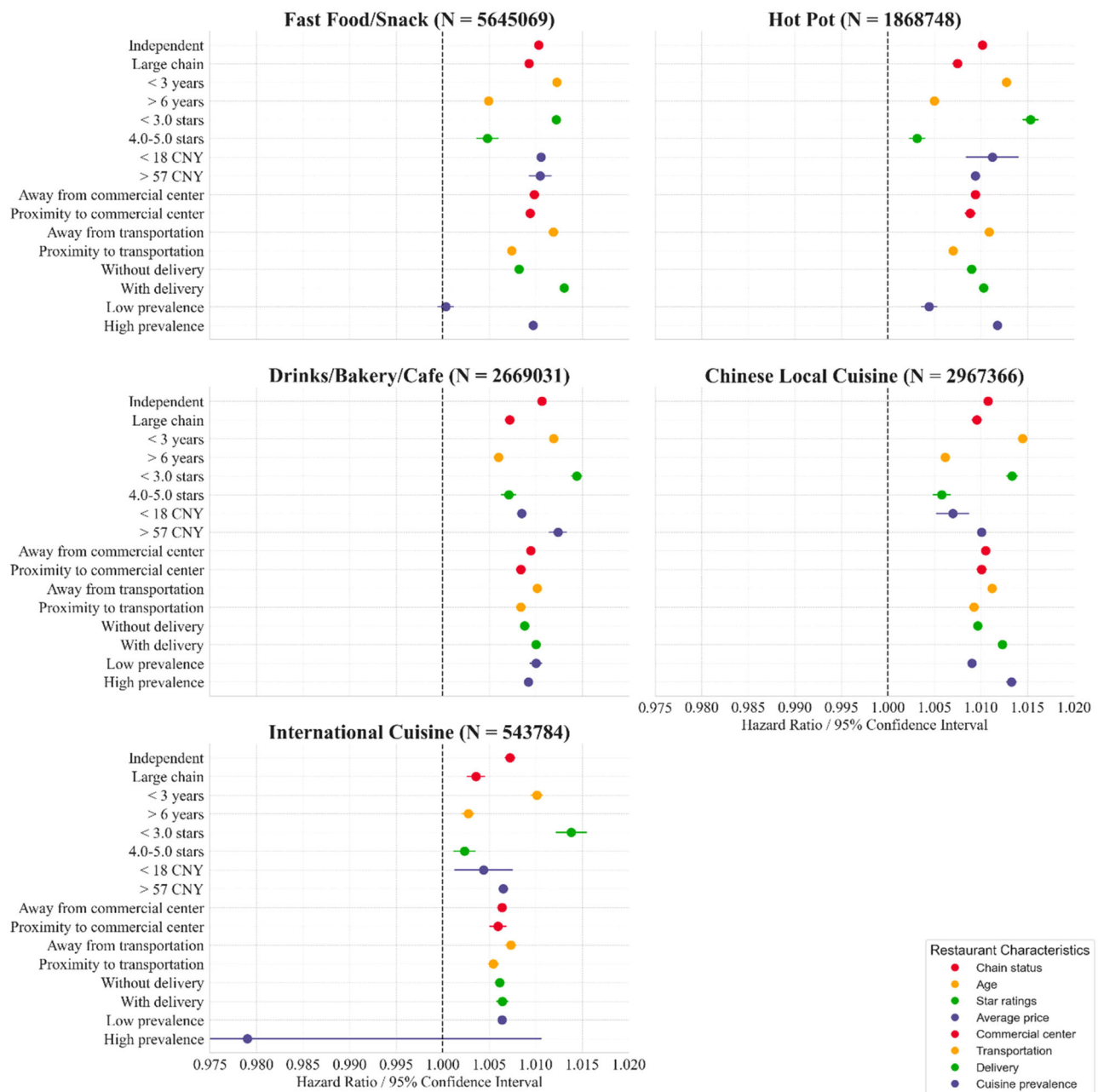


Fig. 6 Impact of restaurant characteristics on closure risk across cuisine categories. This figure illustrates the differential effects of various restaurant characteristics on closure risks within specific cuisine categories. To enhance clarity, we have selected two groups for each restaurant characteristic, such as independent and large chain for chain status, and the lowest and highest groups for cuisine prevalence. “N” refers to the number of observations in each sub-sample. The hazard ratios displayed on the horizontal axis are computed as the exponential of b ($\exp(b)$). A hazard ratio greater than 1 signifies an increased risk of closure, whereas a ratio less than 1 suggests a reduced risk.

Table A7, column 8). This significant difference suggests that restaurants offering more unique or less common cuisines may have better chances of surviving prolonged lockdown. However, this trend varies by cuisine type. For example, high-prevalence establishments specializing in fast food, snacks, hot pots, and barbecues face higher closure hazards, whereas high-prevalence restaurants specializing in international cuisines show negligible effects from lockdown on closure rates (see Fig. 6 and SI Appendix, Table A8).

From the above analysis, it is evident that the star rating of a restaurant contributes to its resilience during lockdown. To delve deeper into restaurant resilience, we explore the heterogeneity effects of other factors in both high- (4-5 stars) and low-rated

(< 3 stars) restaurants in Fig. 7. In high-rated restaurants (see SI Appendix, Table A9), chain establishments, those that have been in operation for many years, and high-prevalence restaurants exhibit higher survival odds. Interestingly, factors such as price, proximity to transportation or commercial centers, and the availability of delivery services do not appear to increase the survival odds in these high-rated establishments. In low-rated restaurants (see SI Appendix, Table A10), older and higher-priced establishments demonstrate higher survival odds. Unlike their higher-rated counterparts, proximity to transportation hubs and delivery services, which typically entail intense competition and higher operational costs, significantly decreases the survival odds for these restaurants. The prevalence of the cuisine type, the

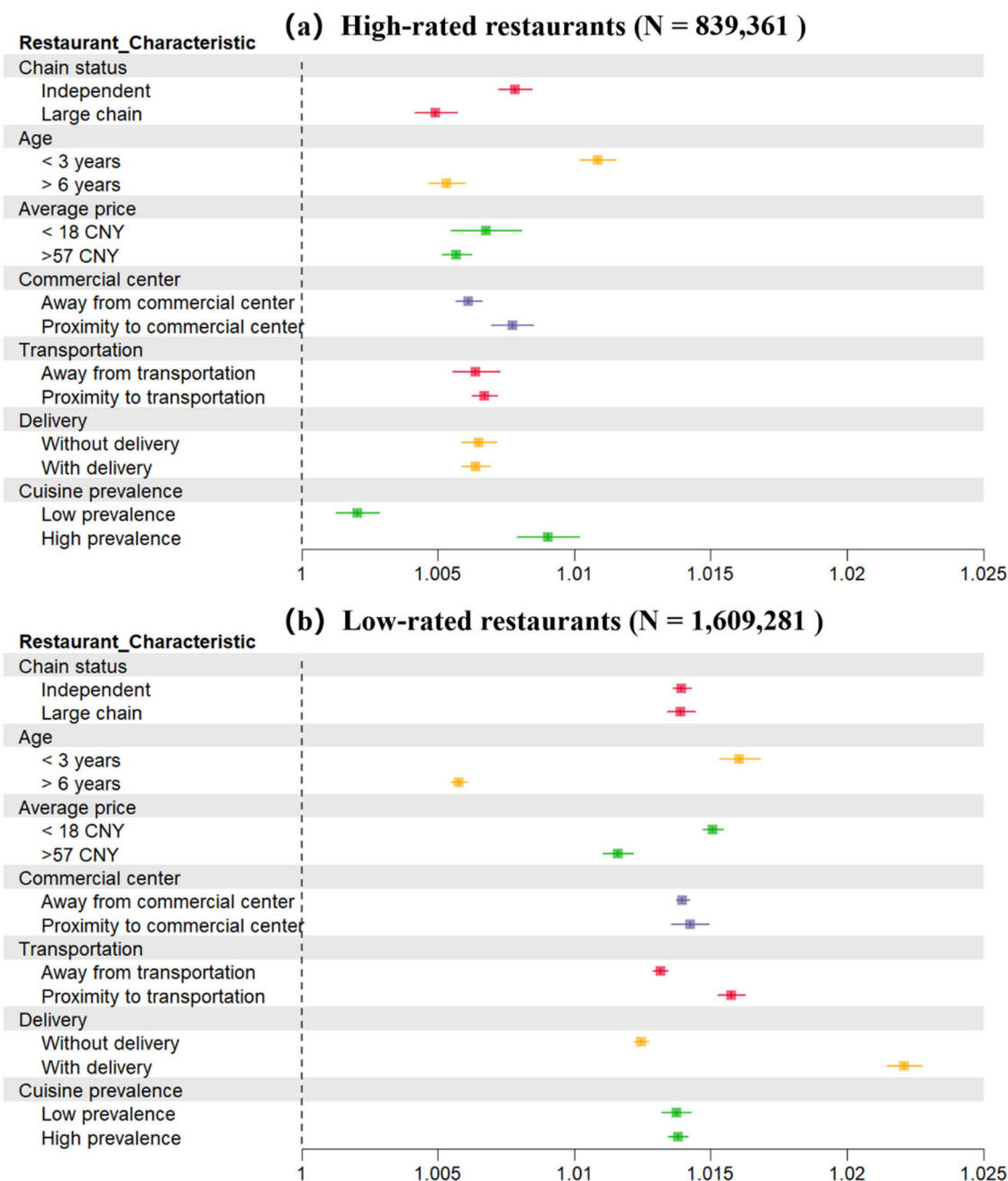


Fig. 7 Impact of restaurant characteristics on closure risk across low- and high-rated restaurants. To enhance clarity, we have selected two groups for each restaurant characteristic, such as independent and large chain for chain status, and the lowest and highest groups for cuisine prevalence. On the horizontal axis, we represent the hazard ratio, which is calculated as the exponential of b (hazard ratio = exp(b)). A hazard ratio greater than 1 indicates an increased risk, while a hazard ratio less than 1 indicates a decreased risk. “N” refers to the number of observations in each sub-sample. Panel (a) presents the hazard ratios for high-rated restaurants (4-5 stars). Panel (b) illustrates the hazard ratios for low-rated (<3 stars) restaurants.

proximity to commercial centers, and whether the restaurant is a chain or independent do not significantly affect survival odds in the lower-rated category.

These findings highlight the complexity of factors that contribute to restaurant resilience during lockdown. The differential impacts observed between high and low-rated restaurants underscore the importance of strategic positioning and unique service offerings, especially in times of crisis. For policymakers and business owners, these insights suggest that while improving quality ratings and cultivating unique characteristics may enhance resilience, attention must also be given to managing costs associated with high-risk factors like delivery and

proximity to major transit points in less favorably rated establishments.

Additional analysis. In our primary analysis, we demonstrate that lockdown duration, with a one-year lag, leads to restaurant closures. However, the impact of lockdown on restaurants may vary over time, resulting in closures either shortly after or persisting over subsequent years. In other words, lockdown measures may lead to restaurant closures in the same year or the following year, and could also have longer-term effects, resulting in closures in the longer time horizon. We introduce three new independent

variables representing restaurant closures within one ($Closure_{it(t \sim t+1)}$), two ($Closure_{it(t \sim t+2)}$), and three years ($Closure_{it(t \sim t+3)}$) (see SI Appendix, Table A4). Using Cox regression at the restaurant level, we reveal that lockdown duration significantly increases the odds of restaurant closures within one year, two years, and three years respectively (see SI Appendix, Table A11). These results reaffirm the significant and lasting negative impact of lockdown measures on restaurant closures, indicating that their effects extend beyond the immediate period to impact establishments in the years ahead.

To assess how different lockdown measures affected the restaurant industry's resilience, we first analyze the impact of the comprehensive lockdown in 2020 on restaurant closures in 2021. Table A12 in the SI Appendix shows that the coefficient is 0.030. Next, we examine the effects of targeted risk zone measures on closures, focusing on data from 2022 and 2023, which reflect the risk zone classifications in 2021 and 2022. The coefficient here is 0.011. Our findings suggest that the comprehensive lockdown had a more significant impact on restaurant survival than the targeted risk zone measures.

Robustness. To assess the robustness of our primary findings, we conduct various robustness checks using alternative specifications (detailed in the SI Appendix, Table A13). We refine our methodology to calculate the direct linear distance between each restaurant and the geographic center of nearby risk zones. Utilizing the Baidu Maps API, we converted the addresses within lockdown zones into precise latitude and longitude coordinates. We then calculated the distance from each restaurant to these points to ascertain proximity. Specifically, we identified restaurants located within 500 m, 1 km, 3 km, and 5 km of a lockdown zone and recorded the number of days they were affected by lockdown measures. Our analysis shows a pronounced increase in the likelihood of closure across all measured distances. This correlation underscores the significant impact of lockdown measures on restaurant sustainability, as detailed in the SI Appendix, Table A14. Additionally, we conduct robustness checks involving using different metrics for independent variables, such as the frequency of pandemic outbreaks, the average number of lockdown days per pandemic outbreak, and the maximum number of consecutive lockdown days annually in restaurant locations. The outcomes from these checks affirm the sensitivity of the results (see SI Appendix Table A15).

Discussion

Utilizing comprehensive restaurant data from Dianping.com, we assess the macro and micro impact of the lockdown on the restaurant sector. The city-level analysis underscores the harsh reality that heightened lockdown severity in a city, in terms of duration and geographic coverage, leads to increased restaurant closures and a reduction in new market entries. Lockdown restricts travel and decreases restaurant foot traffic, impacting revenue streams. The recurring nature of lockdowns escalates operational costs and undermines the confidence of restaurant owners, resulting in a significant number of closures.

Our granular analysis at the restaurant level allows us to identify the characteristics that may buffer restaurants against the adverse effects of lockdown or, conversely, render them more susceptible to closure. First, chain restaurants and older establishments demonstrate relatively lower closure likelihood during lockdown. These findings suggest that standardized operational procedures and accumulated managerial experience, typically present in chain-affiliated or long-standing restaurants, may offer greater organizational stability and resilience under crisis. Additionally, being part of a chain offers more than just operational

advantages; it also serves as a brand signal. A strong brand helps reduce perceived risks by fostering trust, thereby influencing consumer purchase intention.

Second, star ratings and price can serve as quality cues that help consumers make decisions. While the negative impacts of the lockdown are generally consistent across various types of restaurants, those with higher star ratings show significantly more resilience. Star ratings reflect customers' overall evaluations of a restaurant's products and services, and they function as a valuable cue for consumers seeking to assess quality in uncertain situations or when options are limited (Kim et al. 2021a). Such positive evaluations can enhance consumers' perceptions of service quality, increase their intention to visit (Liu et al. 2024), and thereby improve a restaurant's likelihood of remaining operational. Additionally, our findings show that restaurants with higher prices have lower closure likelihood during lockdown. Higher pricing, as another indicator of quality (Rao and Monroe 1989), may help maintain consumer trust and demand, even in difficult times.

Third, restaurants located near commercial centers and public transportation are less likely to close during the pandemic. This is because such locations offer greater accessibility and convenience for consumers, even under lockdown restrictions. Proximity to commercial hubs increases foot traffic, and being near public transportation makes it easier for consumers to visit. The increased visibility and convenience help maintain consumer flow and sales, which are crucial for a restaurant's survival during challenging times. However, the benefits of proximity to commercial centers may be partially offset by intensified local competition, which could limit the protective effect of location advantages.

Finally, we find that offering delivery services and serving common cuisines are associated with a higher risk of closure during lockdown. The effect of delivery services in our study is not consistent with an earlier study based on short-term sales data (Kim et al. 2021b). One possible explanation is that delivery services increase operating costs for smaller or less efficient restaurants, especially due to high platform fees and reduced revenue during lockdown. This effect may be more evident over the long term. Moreover, delivery sales also declined during lockdown (Zhao and Liu 2025), partly because consumers found it difficult to assess food quality when ordering remotely (Kim et al. 2021a). Our additional analysis reveals that restaurants with lower star ratings face a higher risk of closure when offering delivery, suggesting that quality perceptions play a key role in delivery success. These findings underscore the dual-edged nature of delivery services. In addition, we also find that restaurants serving common cuisines are more likely to close. These restaurants face intense competition and offer limited product differentiation, making it harder to retain customers when demand falls.

Our findings highlight the critical role of product and service quality in supporting restaurant survival during crisis periods. High customer ratings, which always reflect consistent quality, safety, and trust, emerge as a key factor in reducing closure risk. This suggests that restaurant owners should prioritize maintaining high service standards and positive customer experiences, which can generate customer loyalty and favorable word of mouth, both crucial during demand shocks. In addition, the results point to the need for careful cost management. Features such as delivery services, while potentially beneficial, may increase operational burdens for smaller or lower-rated restaurants, especially under revenue constraints. Owners should approach third-party platform use with caution and evaluate whether these services align with their capacity and brand positioning.

From a policy perspective, our study underscores the impact of lockdown measures on the restaurant sector, a major component

of the service economy. These findings suggest that blanket restrictions on mobility can unintentionally accelerate business closures in vulnerable industries. Therefore, policymakers should tailor containment measures to local conditions and sector-specific risk, rather than applying uniform lockdown policies. Moreover, targeted support is essential. Policymakers should prioritize assistance to at-risk restaurants, particularly smaller, independent establishments or those operating in highly restricted areas. Such support might include direct subsidies, tax relief, rent deferrals, or training programs that help restaurants transition to lower-contact service models. By recognizing the sector's structural vulnerabilities and responding with tailored interventions, policy efforts can more effectively balance public health goals with economic sustainability.

We acknowledge certain limitations in our study that open avenues for future research. First, our study focuses on restaurant closures as a key sign of business disruption. However, new restaurant openings are also important for understanding the full impact of lockdowns on business activity. Future research could build on our findings by exploring how lockdowns affected entrepreneurs' decisions to start new restaurants and what this means for the industry's recovery over time. Second, our study examines the effects of lockdown on the likelihood of restaurant closure. However, other consequences, such as order numbers, revenue, and consumer behavior, remain unexplored. Future investigations should delve into these aspects to provide a more nuanced understanding of restaurant survival during the pandemic. Third, we collected restaurant data in 2019, just before the COVID-19 outbreak, which serves as a baseline for assessing the impacts of the lockdown on restaurants. However, this limits our ability to compare longer pre- and post-pandemic trends or analyze short-term fluctuations in the restaurant industry. This limitation prevents a more thorough exploration of the crisis's impacts and the effectiveness of governmental interventions. Lastly, our results identify some types of restaurants as being more resilient during lockdown. We suggest that future studies delve into strategies for restaurant recovery post-lockdown, offering additional insights into navigating crises and rebuilding resilience.

Data availability

For access to the research data supporting this study, interested parties may contact the corresponding author.

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Notes

- 1 Based on Meituan's Q3 2023 earnings report, there were approximately 6 million restaurants on the platform. Meituan is China's largest local services platform, similar to Uber Eats or DoorDash for food delivery. Since Meituan and Dianping merged in 2015, their systems have been integrated, and restaurant listings on Dianping are now part of Meituan's overall restaurant database.
- 2 The start date of each full-scale lockdown is typically obtained from official government announcements. However, the end date of lockdown may not always be explicitly stated. In such cases, we assume that the lockdown concludes seven days before the "clearance" day, which marks the first day without any new COVID-19 cases recorded over the past consecutive 14 days. This approach allows us to calculate the duration of full-scale lockdown for each city.
- 3 Given that our restaurant data is collected annually, we use October 31st as the cutoff date for compiling lockdown information for a given year.
- 4 Different local governments classified medium- and high-risk zones differently, leading to varied lockdown measures across these areas. This variation makes it difficult to assess the lockdown impact consistently on specific sites like streets or communities where restaurants may be located. Moreover, medium- and high-risk zones, which may initially cover only a single address or building, could expand to

entire districts or counties if nearby areas face similar restrictions (Chen et al. 2025). This expansion effectively placed the entire district or county under lockdown, affecting any restaurants within those areas (Chen et al. 2025).

- 5 Restaurants listed on Dianping.com are often labeled as being located within or near a mall or shopping center to attract more visitors. Based on this classification, proximity to a commercial center is defined as either (1) a restaurant being situated inside a mall or shopping center or (2) a restaurant being within approximately 1,000 meters of such a location, which is typically within a 10-minute walking distance.
- 6 Restaurants are often labeled as being near public transport facilities, such as subway stations, bus stops, train stations, or airports, on the Dianping website. This label indicates that the restaurant is within 1,500 meters of such facilities. Accordingly, we define proximity to public transportation as a restaurant being located within 1,500 meters of these facilities. If a restaurant does not meet this criterion and lacks the 'near public transport' label on the Dianping website, it is considered not proximate to public transportation.

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Author contributions

SJ and YY contributed to the conceptualization and design of the research. LW and YY conducted the research. SJ was responsible for data collection. LW performed the data analysis. LW, YY, and SJ contributed to the writing and revision of the manuscript. YX provided critical guidance and support throughout the research process.

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Competing interests

The authors declare no competing interests.

Ethical approval

This study does not involve any human participants, personally identifiable information, or sensitive data. All data used in this study were collected from publicly available sources, including government websites and Dianping (a public restaurant review platform). Therefore, ethical approval was not required in accordance with institutional and national regulations.

Informed consent

This study does not involve any human participants or the collection of personally identifiable data. The data used were obtained from publicly accessible sources, and therefore, informed consent was not required.

Additional information

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